

OFFICIAL AGROSYNC TECHNICAL SPECIFICATION

Software Requirements Specification (SRS)

AgroSync Farm Management System

Version: 3.0

SDLC Stage: Production Baseline & Feature Expansion

Status: Approved

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document details the complete functional, non-functional, security, and architectural requirements for **AgroSync** (formerly SmartFarm). It serves as the authoritative single source of truth for engineering, compliance testing, API integration, and operations.

1.2 Scope of the System

AgroSync is an enterprise-grade cloud agricultural management platform designed for modern commercial farms and agribusinesses. The system integrates:

Governance & Financial Control: Executive oversight, multi-category budgeting, and comprehensive Accounts Payable / Accounts Receivable (AP/AR) ledgering for **Farm Administrators**.

Sector Planning & Procurement: Category-level crop/livestock cycle management, supplier procurement, and supervisor task delegation for **Farm Managers**.

Field Operations & Scheduling: Field activity scheduling with automated agronomic presets, harvest logging, input supply consumption, and verified labor task allocation for **Field Supervisors**.

Real-Time Notification Engine: Instant proactive alerts for pending user approvals, low inventory reorder levels, upcoming/overdue tasks, and financial debts.

Unified Transaction Ledger & Pagination: Comprehensive history of sales and supplies with search, type filtering, chronological descending sorting (latest first), and paginated viewing with page size of 5 records.

Labor Compliance & Wage Attribution: Strict validation of employee legal eligibility (\$Age ≥ 18\$) and per-task wage calculation.

1.3 Definitions, Acronyms, and Abbreviations

RBAC: Role-Based Access Control (`ADMIN` , `MANAGER` , `SUPERVISOR`).

PBAC: Privilege-Based Access Control (fine-grained supervisor privileges: `CAN_RECORD_EXPENSES` , `CAN_RECORD_SALES` , `CAN_RECORD_HARVEST` , `CAN_LOG_ACTIVITIES` , `CAN_USE_INVENTORY` , `CAN_VIEW_FINANCIALS`).

Accounts Payable (AP): Cumulative liabilities owed by the farm to input and feed suppliers.

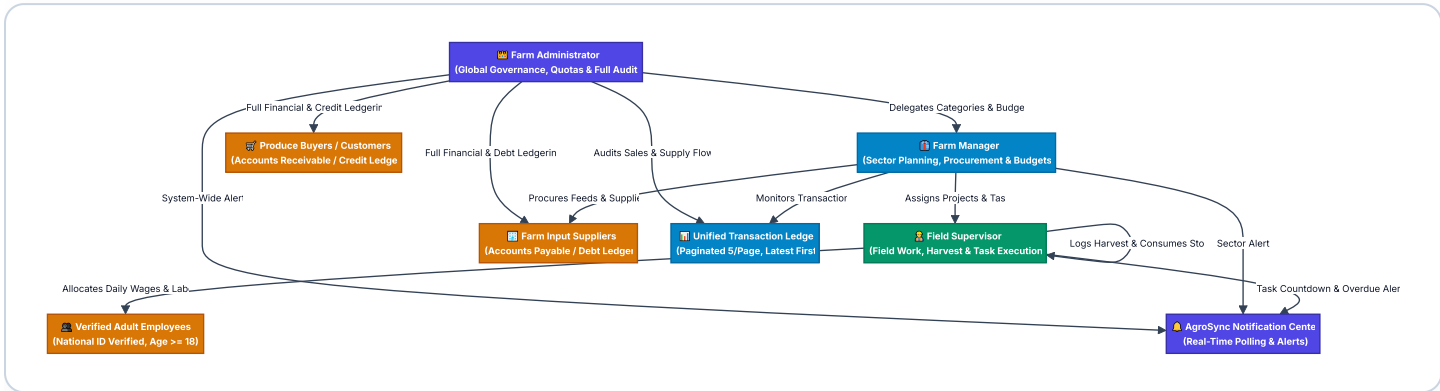
Accounts Receivable (AR): Cumulative uncollected credit balances owed to the farm by produce buyers.

Task Presets: Automated agronomic date calculation offsets (e.g., Planting \$+ 14\$ days \$→\$ Spraying, \$+ 21\$ days \$→\$ Weeding, \$+ 28\$ days \$→\$ Top-Dressing, \$+ 60\$ days \$→\$ Harvesting).

Transaction Pagination: Fixed 5-record paging slicing with latest-first sorting by date descending and secondary ID tie-breaker.

2. Overall Description & System Context

2.1 System Context Diagram



3. Functional Requirements

3.1 Authentication & User Management (FR-AUTH)

FR-AUTH-1 (Bootstrap First Admin): On an empty database (\$0\$ users), the registration system automatically provisions the first registrant as the **Primary Farm Administrator** with active status and full privileges.

FR-AUTH-2 (Approval Lifecycle): Subsequent registrations default to `PENDING_APPROVAL`. Users cannot access farm resources until approved by an Administrator or Manager.

FR-AUTH-3 (Role & Quota Governance):

- Maximum **2 Farm Managers** across the system.
- Maximum **10 Field Supervisors** across the system.

FR-AUTH-4 (Self-Service Password Reset): Email-based token recovery via Resend HTTP API with a 15-minute token expiry window.

3.2 Real-Time Notification Center (FR-NOTIF)

FR-NOTIF-1 (Proactive Aggregation): The system continuously evaluates:

1. `USERS` : Pending user approval requests.
2. `INVENTORY` : Items where `quantityInStock ≤ minStockLevel`.
3. `TASKS` : Overdue farm activities (`$Date < Today$`) and upcoming activities due within 7 days.
4. `FINANCE` : Unpaid supplier purchase invoices (AP) and customer debts (AR).

FR-NOTIF-2 (TopBar Integration): Renders an interactive bell icon with a real-time badge count, pulse animation for new alerts, category filter tabs (`ALL`, `Tasks`, `Stock`, `Users`, `Finance`), and direct navigation links to resolving pages.

FR-NOTIF-3 (Polling Interval): Client refreshes notifications every 45 seconds and on active route transitions.

3.3 Farm Task Scheduling & Presets Engine (FR-SCHED)

FR-SCHED-1 (Quick Agronomic Presets): One-click date and parameter calculation for standard crop cycles:

- `+2 Wks (Spraying)` : Sets scheduled date to `+$14$` days, priority `HIGH`, type `Field Care`.
- `+3 Wks (Weeding)` : Sets scheduled date to `+$21$` days, priority `MEDIUM`, type `Field Care`.
- `+4 Wks (Top-Dressing)` : Sets scheduled date to `+$28$` days, priority `HIGH`, type `Fertilization`.
- `+2 Mos (Harvesting)` : Sets scheduled date to `+$60$` days, priority `URGENT`, type `Harvesting`.

FR-SCHED-2 (Task Lifecycle): Tasks support statuses: `SCHEDULED` , `IN_PROGRESS` , `COMPLETED` , `CANCELLED` and priority tiers: `LOW` , `MEDIUM` , `HIGH` , `URGENT` .

FR-SCHED-3 (Countdown & Overdue Badges): Cards dynamically render badges: `Due Today` *(Amber)*, `In X days` *(Blue)*, `Overdue by X days` *(Red)*, `Completed` *(Green checkmark)*.

FR-SCHED-4 (One-Click Completion): Allows users to mark tasks as completed directly from the activity feed via `PATCH /activities/{id}/status` .

3.4 Production, Projects & Inventory Management (FR-PROD)

FR-PROD-1 (Category & Project Hierarchy): Category (e.g. Dairy, Poultry, Horticulture) $\rightarrow$ Projects (e.g. Greenhouse 1, Broiler Batch A) $\rightarrow$ Records (Expenses, Activities, Harvest, Sales).

FR-PROD-2 (Inventory & Supply Consumption): Direct tracking of chemical, feed, fertilizer, and tool stocks with stock-deduction on farm usage (`POST /inventory/{id}/use`).

FR-PROD-3 (Harvest & Produce Stock Control): Sales cannot exceed harvested stock for any produce item.

3.5 Trade, Debt Ledgering & Financial Analytics (FR-FIN)

FR-FIN-1 (Accounts Payable / Suppliers): Tracks external suppliers, purchase invoices, payment modes (`CASH` , `BANK` , `MPESA` , `CREDIT_LEDGER`), and outstanding liabilities.

FR-FIN-2 (Accounts Receivable / Customers): Tracks customer debt, invoice balances, payment histories, and credit limits.

FR-FIN-3 (Executive Dashboard Metrics): Multi-card auto-fit grid displaying:

- Total Revenue Collected
- Revenue Received (Cash In Bank)
- Cumulative Customer Debt (Uncollected AR)
- Unpaid Supplier Invoices (Pending AP)
- Total Expenses & Net Farm Value

FR-FIN-4 (Unified Transaction History & Pagination Engine):

- Consolidates incoming crop/livestock sales and outgoing farm supply procurement records into a unified chronological ledger.
- **Ordering:** Guarantees strict `latest first` presentation by sorting on transaction date descending (`dateB.localeCompare(dateA)`), with a deterministic tie-breaker on ID descending.
- **Pagination:** Fixed page size of **5 records per page** (`TX_PAGE_SIZE = 5`). Provides intuitive pagination controls (Previous/Next buttons, active numeric page pills, and record count indicators `Showing X to Y of Z transactions`).
- **Filtering & Search:** Real-time text search across reference code, party name, and item description, combined with quick type filtering (`All` , `Sales Only` , `Supplies Only`). Page resets to 1 automatically upon filter or search change.

3.6 Labor Compliance & Wage Attribution (FR-LABOR)

FR-LABOR-1 (National ID & Age Enforcement): Strict age validation (`$Age >= 18$`) based on date of birth and verified government ID numbers.

FR-LABOR-2 (Task Rostering & Wage Calculation): Assignment of multiple verified employees to specific activities with custom wage amounts and payment status tracking.

4. Non-Functional Requirements (NFR)

4.1 Performance & Scalability

NFR-PERF-1: Sub-100ms average backend response time for dashboard aggregations and notification polling.

NFR-PERF-2: Database connection pool optimization with `spring.jpa.open-in-view=false`.

NFR-PERF-3: Frontend asset bundle optimization with code splitting, compressed SVG assets, and sub-3s production builds.

4.2 Security & Access Control

NFR-SEC-1: Passwords encrypted using BCrypt (strength 10).

NFR-SEC-2: Stateless request validation with `X-User-Id` and `X-User-Role` verification.

NFR-SEC-3: Supervisor financial isolation (cost amounts, sales margins, and company cash balances are masked for non-authorized supervisors).

4.3 Reliability & Availability

NFR-REL-1: Graceful zero-state handling when network disruptions occur.

NFR-REL-2: Automated database schema migration and validation on application startup.